MILITARY SPECIFICATION

APPLESAUCE, SEMI-SOLID, THERMOSTABILIZED TUBE PACK

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers tube packed semi-solid applesauce as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-44320 - Packaging and Processing of Foods in Collapsible Containers (Tubes)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8915

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Canned Applesauce

(Application for copies should be addressed to the Standardization Section, Fresh Products Branch, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Washington, DC 20090-6456.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, Environmental Protection Agency, WH550D 401 M. Street, S.W., Washington, DC 20460.)

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2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Applications for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Sugar, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet, or a combination of both and shall comply with MIL-STD-900.

3.2.2 Citric acid. Citric acid shall comply with the Food Chemicals Codex.

3.2.3 Malic acid. Malic acid shall comply with the Food Chemicals Codex.

3.2.4 Applesauce, canned. The applesauce shall be sweetened, natural color and flavor type, and regular (comminuted) style. The canned applesauce shall be U.S. Grade A of the U.S. Standards for Grades of Canned Applesauce and shall be of the latest season's pack.

3.2.5 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and processing. Preparation shall be on a continuous basis.

3.3.1 Product preparation. The product shall be prepared as follows:

a. The canned applesauce shall be acidified, as necessary, to a pH of 3.6 to 3.8 by the addition of food grade malic or citric acid.

b. The applesauce shall be heated to 165° to 180°F in a steam jacketed kettle, with care being taken to avoid scorching of the product.

c. Granulated sugar shall be added only when necessary to assure compliance with the finished product Brix requirement (see 3.5).

3.4 Filling and sealing. Five ounces of product shall be filled into a tube as specified in MIL-P-44320 immediately after heating. The temperature of the product at the time of filling shall be in the range of 165° to 170°F. The filled tubes shall be immediately given a saddle-back fold and crimped securely, then sealed under vacuum by flushing the head space of each filled tube with super-heated steam. The filled and sealed tubes shall be pasteurized within 2 hours after sealing in accordance with MIL-P-44320.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, moldy, or sulfurous.

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c. The product shall be tan in color.

d. The product shall have a characteristic applesauce flavor.

e. The product shall not be rubbery.

f. The product shall be smooth, without lumps, and shall not be excessively thick or thin.

g. The average net weight shall be not less than 4.8 ounces.

h. No individual tube shall contain less than 4.7 ounces of product.

i. The pH shall be not less than 3.60 or more than 3.80 (see 4.5.6.1).

j. The Brix value shall be not less than 20° nor more than 22° (see 4.5.6.2).

j. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.5.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample in palatability and overall appearance (see 4.4, 6.1, and 6.2).

3.6 Plant qualifications. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods," and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

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4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed tube inspection. Inspection of filled and sealed tubes shall be in accordance with the quality assurance provisions of MIL-P-44320.

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4.5.4 Net weight inspection. The net weight of the filled and sealed tubes shall be determined by weighing each sample unit on a suitable scale tared with a representative empty tube. Any individual net weight of less than 4.7 ounces shall be scored as a minor defect. The lot size shall be expressed in units of tubes. The sample unit shall be one filled and sealed tube. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 4.8 ounces.

4.5.5 Product inspection. The filled and sealed sample tubes shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the retort process. The tubes shall be opened and inspected for the defects indicated in table I. The lot size shall be expressed in tubes. The sample unit shall be the contents of one tube. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	
101	Product is rubbery
102	Product is not smooth
103	Product is excessively thick or thin
104	Product is lumpy
105	Product shows evidence of excessive heating (materially darkened or scorched)
106	Product does not have a characteristic applesauce flavor
107	Product does not have a characteristic tan color

1/ The presence of foreign material (e.g., dirt, insects, insect parts, hair, wood, paper, paint, glass, metal), foreign odor or flavor (e.g., burnt, scorched, stale, sour, rancid, musty, moldy, sulfurous), or foreign color shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparision shall be performed only when deemed necessary by an Agricultural Marketing Services (AMS) agent.)

4.5.6 Product testing. The filled and sealed tubes shall be tested for pH as specified in 4.5.6.1 and for degrees Brix as specified in 4.5.6.2.

4.5.6.1 pH testing. Thirteen tubes of product shall be selected at random from the lot and individually tested for pH in accordance with the applicable chapter for pH determination of the Official Methods of Analysis of the Association of Official Analytical Chemists. The results shall be reported to the nearest 0.01. Any pH less than 3.60 or more than 3.80 is a major defect and shall be cause for rejection of the lot.

4.5.6.2 Degrees Brix. Thirteen tubes of product shall be selected at random from the lot and individually tested for degrees Brix. Degrees Brix of finished product, corrected to 20°C, shall be determined by means of a refractometer, whether directly with the use of a sugar refractometer or correlated with refractive indices of sucrose solutions at 20°C, if the readings are in terms of refractive index. No correction shall be made for insoluble solids or acid content. If it is difficult to obtain a clear reading, the product should be filtered through a milk-filter pad. Any Brix reading less than 20° or more than 22° is a major defect and shall be cause for rejection of the lot.

4.5.7 Packaging inspection. Packaging inspection shall be in accordance with the quality assurance provisions of MIL-P-44320.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44320.

6. NOTES

6.1 Ordering data. Acquistion documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is specified (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction sample (see 3.5.1 and 6.2).

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6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Subject term (key word) listing.

Applesauce
Semi-solid
Tube

Custodian:

Army - GL

Review activities:

Army - MD, TS
Navy - MC, SA
Air Force - 50
DP-SS

Preparing activity:

Army - GL

Project No. 8940-A592

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-44321(GL)	Applesauce, Semi-Solid, Thermostabilized Tube Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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